photoelectric current is set up in the circuit which is measured using micro ammeter.

The variation of photocurrent with respect to (i) intensity of incident light (ii) the potential difference between the electrodes (iii) the nature of the material and (iv) frequency of incident light can be studied with the help of this arrangement.

Effect of intensity of incident light on photoelectric current

To study the effect of intensity of incident light on photoelectric current, the frequency of the incident light and the accelerating potential V of the anode are kept constant. Here the potential of A is kept positive with respect to that of C so that the electrons emitted from C are attracted towards A. Now, the intensity of the incident light is varied and the corresponding photoelectric current is measured.

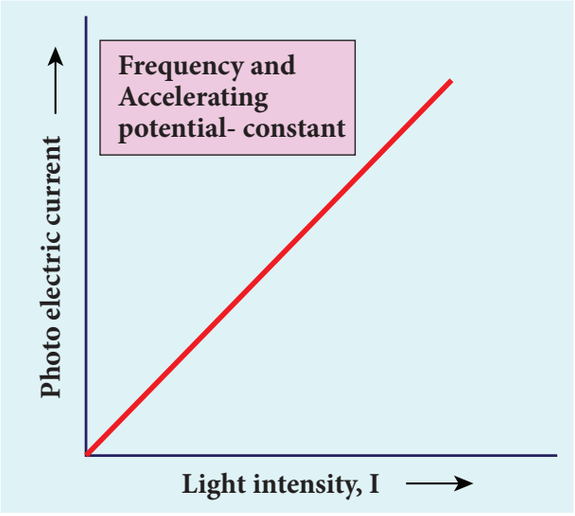


Figure 8.9 Variation of photocurrent with intensity

A graph is drawn between light intensity along x-axis and the photocurrent along y-axis. From the graph in Figure 8.9, it is evident that *photocurrent – the number of electrons emitted per second – is directly proportional to the intensity of the incident light.*



Here, intensity of light means brightness. A bright light has more intensity than a dim light.

8.2.3 Effect of potential difference on photoelectric current

To study the effect of potential difference V between the electrodes on photoelectric current, the frequency and intensity of the incident light are kept constant. Initially the potential of A is kept positive with respect to C and the cathode is irradiated with the given light.

Now, the potential of A is increased and the corresponding photocurrent is noted. As the potential of A is increased, photocurrent also increases. However a stage is reached where photocurrent reaches a saturation value (saturation current) at which all the photoelectrons from C are collected by A. This is represented by the flat portion of the graph between potential of A and photocurrent (Figure 8.10).

When a negative (retarding) potential is applied to A with respect to C, the current does not immediately drop to zero because the photoelectrons are emitted with some definite and different kinetic energies. The kinetic energy of some of the photoelectrons is such that they could overcome the retarding electric field and reach the electrode A.

When the negative (retarding) potential of A is gradually increased, the photocurrent starts to decrease because more and more photoelectrons are being repelled away from reaching the electrode A. The photocurrent becomes zero at a

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